



Practical Type Inference with Levels

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Modern functional languages rely on sophisticated type inference algorithms. However, there often exists a gap between the theoretical presentation of these algorithms and their practical implementations. Specifically, implementations employ techniques not explicitly included in formal specifications, causing undesirable consequences. First, this leads to confusion and unforeseen challenges for developers adhering to the formal specification. Moreover, theoretical guarantees established for a formal presentation may not directly translate to the implementation. This paper focuses on formalizing one such technique, known as *levels*, which is widely used in practice but whose theoretical treatment remains largely understudied. We present the first comprehensive formalization of levels and demonstrate their applicability to type inference implementations.

CCS Concepts: • **Software and its engineering** → **Functional languages**; *Polymorphism*.

Additional Key Words and Phrases: type inference, levels, higher-rank polymorphism, type regions

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1 Introduction

Modern functional programming languages utilize sophisticated type inference mechanisms derived from the Hindley-Milner algorithm [Damas and Milner 1982; Hindley 1969] to support expressive type systems. These expressive types include *higher-rank polymorphism* [Dunfield and Krishnaswami 2013; Odersky and Läufer 1996; Peyton Jones et al. 2007], *impredicativity* [Emrich et al. 2020; Parreaux et al. 2024; Serrano et al. 2020], *higher-kinded types* [Xie et al. 2019], and *existential types* [Eisenberg et al. 2021; Läufer and Odersky 1992].

Most studies in type inference often involve both a formal declarative specification as well as a corresponding algorithmic type system. A central focus of such work lies in establishing *soundness* and *completeness* of the algorithmic system, demonstrating that the algorithmic system faithfully captures the properties of the declarative specification.

However, while soundness and completeness are indeed fundamental for type inference algorithms, practical implementations demand more than theoretical guarantees. A crucial aspect often overlooked in formal presentations is the actual *implementation techniques* in modern languages. These techniques are important as practical type inference systems must not only be sound, but also performant, principled, and easy-to-maintain to address practical concerns such as efficiency and code clarity.

As an example, consider the following program:

let $f = \lambda x \rightarrow x$ **in** (f 1, f *True*)

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Hindley-Milner systems apply *let generalization* to infer the polymorphic type $\forall a. a \rightarrow a$ for f . However, standard presentations of let generalization [Damas and Milner 1982; Hindley 1969; Peyton Jones et al. 2007, 2006] involve traversing the entire typing context to decide the free type variables in the type of $\lambda x. x$. This traversal can be inefficient in larger programs with numerous variables and nested let expressions. Practical implementations often use more efficient generalization strategies.

The omission of implementation techniques in the formal presentations of type inference algorithms has undesirable consequences. First, it creates a gap between the theoretical description and the practical realization of these algorithms. Consequently, developers who implement algorithms based solely on formal specifications may encounter unforeseen performance bottlenecks or end up implementing additional ad-hoc checks, and only later discover more practical implementation strategies. Moreover, and perhaps more importantly, theoretical guarantees established for a formal presentation may not directly translate to the implementation. This discrepancy can undermine the reliability and predictability of type inference algorithms.

Therefore, we argue that it is essential to bridge this gap by incorporating the key implementation insights into presentations of type inference algorithms. This involves presenting the essential concepts of implementation techniques without delving into every low-level detail, as including every implementation nuance can easily lead to a cluttered and unwieldy presentation. An overly prescriptive approach is also impractical, as developers may still make varied choices regarding concrete implementation details.

To this end, this paper focuses on *levels*, a technique widely used in practical type inference implementations, but whose formal treatment remains largely understudied. Originally proposed by Rémy [1992], levels have been employed in various type checkers, particularly for OCaml and Haskell, to effectively implement features including let-generalization, escape checking of skolems in higher-rank polymorphic systems, and type regions, and more. Surprisingly, despite their prevalence, a formalism of levels remains largely absent from the presentations of those algorithms. Notable exceptions include the original formalism by Rémy [1992] and subsequent work by Kuan and MacQueen [2007], which focused only on levels for let generalization.

This paper aims to address this gap by providing the first comprehensive formalism of levels beyond let generalization, and demonstrate their broader applications within type inference implementations. The formalization provides a novel theoretical foundation for levels, clarifying their role and interactions, particularly when used for multiple purposes within a type inference algorithm. Moreover, we establish desirable properties including soundness and completeness and reveal novel level-related properties, ensuring the reliability of level-based type inference. We believe this precise and formal account of levels will benefit both practitioners and researchers in the field by providing a clearer understanding of levels and their applications.

We offer the following contributions:

- We provide the first formalism of a type system that incorporates let generalization, higher-rank polymorphism, and local data types within a level-based framework.
- We prove that the level-based declarative system (§4) is sound and complete with respect to a non-level-based declarative system (§3). These proofs have been mechanized using the Coq proof assistant [Coq Team 2024], demonstrating key level-related properties and invariants, which, to the best of our knowledge, have not been previously established (§5).
- We present a level-based algorithmic type system, featuring a novel *polymorphic promotion* process for resolving level constraints. We prove the algorithm to be sound and complete with respect to the level-based declarative type system (§6).

- We have implemented and evaluated the level-based type inference algorithm in the Koka compiler (§7), a strongly typed functional language with a polymorphic type-and-effect system.
- We explore language extensions and show how levels are used to support them within modern type checkers such as GHC and the OCaml type checker (§8).

Our formalism is detailed, and some rules are elided for space reasons. The complete set of rules, as well as proofs of stated theorems for the algorithmic type system are included in the appendix.

2 Overview

This section gives an overview of our work; we use Haskell-like syntax for examples.

2.1 Hindley Milner and Let Generalization

The Hindley-Milner (HM) type system [Damas and Milner 1982; Hindley 1969] provides a foundation for many modern type inference algorithms. A key feature of HM is its ability to incorporate parametric polymorphism while still being able to infer the most general type (i.e. *the principal type*) of a program without requiring user-provided annotations.

As an example, consider the following program:

```
let f = λx → x in (f 1, f True)  -- f : ∀a. a → a
```

Here, f is applied to arguments of two different types, *Int* and *Bool* respectively. Fortunately, since the variable x in the expression $\lambda x. x$ is unconstrained, its type can be generalized. This allows HM to infer a polymorphic type $\forall a. a \rightarrow a$ for f , ensuring that the program successfully type-checks.

However, generalization must be handled with care. Specifically, consider:

```
λx → let y = x in (y + 1, not y)  -- error
```

In this case, the definition of y refers to x . While it might seem that x is unconstrained within the definition of y , leading to a tempting generalization of y 's type to $\forall a. a$, this would be incorrect! Rather, since x is defined outside of y 's definition, we cannot generalize over its type.

To correctly implement generalization, the HM let generalization is formalized as follows:

$$\frac{\Psi \vdash e_1 : \tau_1 \quad \Psi, x : \forall \bar{a}. \tau_1 \vdash e_2 : \tau_2 \quad \bar{a} \notin \text{ftv}(\Psi)}{\Psi \vdash \text{let } x = e_1 \text{ in } e_2 : \tau_2} \text{ HM-LET}$$

The rule first infers the type of e_1 , getting τ_1 . It then generalizes τ_1 to $\forall \bar{a}. \tau_1$ as the type of x , and adds x to the typing context to infer the type of e_2 , getting τ_2 . Importantly, the side condition $\bar{a} \notin \text{ftv}(\Psi)$ requires the generalized type variables to not appear in the free type variables of Ψ .

To illustrate the importance of the side condition in rule **HM-LET**, let us revisit our previous examples. In the first case, we have $\bullet \vdash \lambda x. x : a \rightarrow a$, allowing us to generalize the type to obtain $f : \forall a. a \rightarrow a$. However, in the second case, we have $x : a \vdash x : a$, and the occurrence of a in the typing context prevents generalization, resulting in $y : a$, thus correctly rejecting the program. A language implementor for the HM type system will then use the algorithmic version of this rule, which takes $\bar{a} = \text{ftv}(\tau_1) - \text{ftv}(\Psi)$, explicitly calculating the set of type variables to generalize.

However, implementing generalization directly this way can lead to inefficiencies. Specifically, each generalization step requires traversing the entire typing context ($\text{ftv}(\Psi)$) to determine the free type variables. This traversal can become computationally expensive, especially when dealing with larger contexts containing numerous definitions.

To address such inefficiency, we employ the following let generalization rule **LET**:

$$\frac{\Psi \vdash^{n+1} e_1 : \tau_1 \quad \Psi, x : \forall \text{ftv}^{n+1}(\tau_1). \tau_1 \vdash^n e_2 : \tau_2}{\Psi \vdash^n \text{let } x = e_1 \text{ in } e_2 : \tau_2} \text{ LET} \qquad \frac{\Psi, x : \tau_3^{\leq n} \vdash^n e : \tau_4}{\Psi \vdash^n \lambda x. e : \tau_3 \rightarrow \tau_4} \text{ LAM}$$

Notably, the typing judgment is now indexed by an integer n , called a *level*. This level is incremented when typing the expression e_1 , effectively tracking the nesting depth of let expressions. (The concept of levels extends beyond nested lets, as we will explore later.) Moreover, each type variable is now also associated with a level. Importantly, a type variable can only be used if its level is less than or equal to the current typing level. This invariant is maintained throughout the type inference process. In particular, when typing a lambda expression (rule **LAM**), the type of the argument x is required to have a level at most n . As a result, the typing context Ψ in rule **LET** only contains variables at a level at most n , while the type τ_1 may include variables at level $n + 1$. Upon exiting e_1 , any variables at level $n + 1$ are guaranteed to not occur in Ψ . Therefore, we can generalize those variables in τ_1 .

We can see that rule **LET**, compared to rule **HM-LET**, offers a more efficient approach to generalization. Specifically, rule **LET** calls ftv^{n+1} , which traverses the type τ_1 to identify free type variables at level $n + 1$, rather than traversing the typing context. A formalism with rule **LET** was first introduced by Rémy [1992]¹, which has inspired various practical implementations. Rémy's work focused only on let generalization. In contrast, this work demonstrates the broader applicability of levels to other type features. Moreover, we support generalization in a bidirectional type system. Furthermore, while the declarative specification of levels assumes an implicit mapping from variables to their levels, we additionally provide a mechanization of the level-based system, making such mapping explicit and establishing key invariants and properties for a more rigorous treatment.

2.2 Levels for Higher-Rank Polymorphism

Higher-rank polymorphism allows universal quantifiers to appear nested. Consider the following program taken from Peyton Jones et al. [2007]:

```
f :: (∀a. [a] → [a]) → ([Bool], [Char])
f x = (x [True, False], x ['a', 'b'])
```

Here, f takes a polymorphic function as an argument, making f itself a rank-2 function. The argument x can thus be applied to different list types. As an example, $f \text{ reverse}$ is a valid application, where reverse takes a list and returns it in the reverse order.

However, care needs to be taken when type-checking higher-rank polymorphic programs. In particular, assuming $(g : \forall a b. a \rightarrow b \rightarrow b)$, consider the following program:

```
(λ(f :: ∀c. c → ∀d. d → d) → f 1) g -- error
```

Here the function expects an argument of type $(\forall c. c \rightarrow \forall d. d \rightarrow d)$, while g has type $(\forall a b. a \rightarrow b \rightarrow b)$. This requires us to check a subtyping constraint $(\forall a b. a \rightarrow b \rightarrow b) <: (\forall c. c \rightarrow \forall d. d \rightarrow d)$. However, such a subtyping relation does not hold [Dunfield and Krishnaswami 2013; Odersky and Läufer 1996]. To illustrate why, let's try to resolve the constraint. First, we skolemize c , by removing the universal quantifier and replacing the bound type variable with a fresh skolem variable c . We can then instantiate a with c . However, we need to instantiate b before skolemizing d . Consequently, d falls outside the scope of b , preventing the subtyping relation from holding.

In an implementation, the system will first instantiate b with a unification variable standing for its instantiation type to be solved, and then skolemize d . The system must then ensure that the unification variable cannot be solved as the skolem d . This highlights a crucial aspect of higher-rank type system: the importance of managing the relative scope of unification and skolem variables.

Dunfield and Krishnaswami [2013] present an elegant formalism of higher-rank polymorphism based on *ordered contexts* [Gundry et al. 2010]. This approach carefully tracks the relative scope of unification and skolem variables by imposing a strict ordering of elements in the context, and a

¹Rémy [1992] used the term *ranks* for the integer n , while modern type checkers generally refer to it as a *level*.

unification variable can only be solved with variables preceding it. This ensures well-scopedness, but maintaining such an ordered context can introduce significant overhead in practical implementations. Peyton Jones et al. [2007] ensure correctness by incorporating additional checks in the algorithmic type system. Specifically, writing σ for polymorphic types, when checking $\sigma_1 <: \forall a. \sigma_2$, the implementation skolemizes a by replacing it with a fresh skolem variable b , and recursively checks $\sigma_1 <: \sigma_2[a := b]$, producing a substitution S from unification variables to types. The system then checks that $b \notin \text{ftv}(S(\sigma_1))$ and $b \notin \text{ftv}(S(\forall a. \sigma_2))$, successfully preventing skolems from escaping their scope through unification variables after applying the substitution S . However, ensuring that an implementation has incorporated complete and sufficient checks (for skolem escape or beyond) can be a rather subtle matter.

In our system, we demonstrate that levels can effectively implement skolem escape checks. The key idea is to associate each skolem variable with a level. In particular, upon entering the scope of a skolem, such as checking $\sigma_1 <: \forall a. \sigma_2$ and skolemizing the type on the right, we increment the typing level, and associate the skolem a with this new level. Since unification variables in σ_1 have lower levels, they cannot be unified with skolems at higher levels, preventing skolems from escaping. This way, skolem escape checks are now implemented in the same level-based framework.

Notably, levels now start serving multiple purposes. Since subtyping can also increment the level, levels no longer always correspond to the nesting depth of lets. Moreover, subtyping can introduce variables with levels higher than those previously used when entering the scope of let expressions. This seems to suggest that the generalization in rule **LET** should be updated from $\text{ftv}^{n+1}(\cdot)$ to $\text{ftv}^{\geq n+1}(\cdot)$, to include all variables with levels greater than or equal to $n + 1$. Surprisingly, we show that in our system the generalization over $\text{ftv}^{n+1}(\cdot)$ remains sound and complete. This subtle nuance stresses again the importance of a rigorous formal analysis of levels.

2.3 Type Regions

Let us now turn our attention to type regions, specifically focusing on local datatype declarations.² As an example, the following program declares a datatype *Tree* with a scope limited to the region following the **in** keyword:³

```
data Tree = Leaf Int | Node Tree Tree in
  let f x = case x of Leaf i → i; Node y z → f y + f z
  in f (Node (Leaf 2) (Leaf 3)) -- 5
```

Importantly, the type *Tree* cannot escape its declared scope. The following program will get rejected:

```
data Tree = Leaf Int | Node Tree Tree in
  Leaf 5 -- error
```

This restricted scope exemplifies the concept of type regions, similar to type declarations within local modules (as in OCaml) or type variables unpacked from existential types. To enforce this restriction, the type system must ensure that *Tree* does not appear in the return type of the expression following the declaration. This can be achieved through a straightforward syntactic check of the return type.

Interestingly, we can also leverage levels to implement this scope restriction. Specifically, when entering the scope of a type region, we increment the current typing level, and associate *Tree* with this new level. Upon exiting the scope, we check that the return type has a level less than or equal to the previous level, which effectively ensures that *Tree* does not occur free in the return type. While obtaining the level of a type might involve traversing the entire type structure, leading to a

²The datatype declarations here correspond to ML/Haskell-style *generative* datatypes [MacQueen et al. 2020, §4.3.3].

³While we use Haskell-like syntax for illustrative purposes, Haskell does not support local datatype declarations.

cost similar to directly searching for *Tree*, this approach highlights the versatility of a level-based framework. Moreover, checking the level of a type can also be implemented through efficient lookup mechanisms, especially in systems that explicitly track the level of an entire type (§8).

2.4 This Work

In this work, we present a novel type system formalism combining let generalization, higher-rank polymorphism, and local datatype declarations in a unified level-based framework. This showcases the versatility of levels in type inference, enabling programmers to implement these different features through a common mechanism. Why choose this particular combination of features? Because they demonstrate the key roles levels play in modern type checkers, for generalization, subtyping and unification, and scope checking, respectively. These features also illustrate the interplay of levels when serving multiple purposes within an implementation. We demonstrate how the notion of levels can be further applied to other language extensions and how they are implemented in modern type checkers in §8.

In the rest of this paper, we begin by presenting a non-level-based declarative system, and then prove that our level-based system is sound and complete with respect to the non-level-based specification. These proofs have been mechanized to capture the subtleties of the calculus. We then present a corresponding level-based type inference algorithm. Our formalisms establish various key level-related properties, providing a foundation for practical type inference with levels.

3 Declarative Type System

This section presents a declarative higher-rank polymorphic type system without levels, similar to the one in Dunfield and Krishnaswami [2013]; Peyton Jones et al. [2007], extended with local datatype declarations. This system serves as the base system. In next section we will introduce a level-based system and then establish its soundness and completeness with respect to this system.

3.1 Syntax

Fig. 1 presents the syntax of expressions and types used in this section and §4. Expressions e include literals i , variables x , lambdas $\lambda x. e$, annotated lambdas $\lambda x : \sigma. e$, applications $e_1 e_2$, annotated expressions $e : \sigma$, let expressions **let** $x = e_1$ **in** e_2 , and local datatypes **data** $T = \overline{D_i \sigma_j^i}^j$ **in** e . For simplicity, we focus on datatypes without type parameters.⁴ We assume type annotations (in $e : \sigma$ and $\lambda x : \sigma. e$) are user-provided and thus are always closed (i.e. without free type variables).

Polymorphic types σ include universally quantified types $\forall a. \sigma$, functions $\sigma_1 \rightarrow \sigma_2$, and monotypes τ . Monomorphic types τ contain no universal quantifiers, and include the integer type Int , type variable a , monomorphic functions $\tau_1 \rightarrow \tau_2$, and datatype T .

Type contexts Ψ track the type of variables, the datatypes, and the types of data constructors.

3.2 Typing

Fig. 2 presents the bidirectional typing rules. For space reasons, we show only selected rules; the complete set of rules can be found in the appendix. The typing judgment has two modes: type inference $\Psi \vdash e \Rightarrow \sigma$ infers the type σ of e , while type checking $\Psi \vdash e \Leftarrow \sigma$ checks e against a given type σ .

Rule **T-LAM** non-deterministically guesses a monotype τ for variable x , and adds $x : \tau$ to the context to type-check the body e . Rule **T-APP** first infers the type of e_1 , getting σ . Since σ must be a function type, the rule uses the *matching* judgment $\triangleright$ to match the type into a function type,

⁴We foresee no fundamental challenges in supporting parameterized data types, which primarily entails incorporating higher-kinded types. Other advanced datatype features (e.g. GADTs) would require further extensions (§8).

expression $e ::= i \mid x \mid D \mid \lambda x. e \mid \lambda x : \sigma. e \mid \overline{e_1 e_2} \mid e : \sigma$
 $\mid \text{let } x = e_1 \text{ in } e_2 \mid \text{data } T = \overline{D_i \overline{\sigma_j^j}^i} \text{ in } e \quad (\sigma \text{ closed})$
 polytype $\sigma ::= \forall a. \sigma \mid \sigma_1 \rightarrow \sigma_2 \mid \tau$
 monotype $\tau ::= \text{Int} \mid a \mid \tau_1 \rightarrow \tau_2 \mid T$
 context $\Psi ::= \bullet \mid \Psi, x : \sigma \mid \Psi, T \mid \Psi, D : \sigma$

Fig. 1. Syntax

$\Psi \vdash e \Rightarrow \sigma$	(Type Inference)		
$\frac{\text{T-LAM} \quad \Psi, x : \tau \vdash e \Rightarrow \sigma}{\Psi \vdash \lambda x. e \Rightarrow \tau \rightarrow \sigma}$	$\frac{\text{T-APP} \quad \Psi \vdash e_1 \Rightarrow \sigma \quad \sigma \triangleright \sigma_1 \rightarrow \sigma_2 \quad \Psi \vdash e_2 \Leftarrow \sigma_1}{\Psi \vdash e_1 e_2 \Rightarrow \sigma_2}$	$\frac{\text{T-ANNO} \quad \Psi \vdash \sigma \quad \Psi \vdash e \Leftarrow \sigma}{\Psi \vdash e : \sigma \Rightarrow \sigma}$	
$\frac{\text{T-LET} \quad \Psi \vdash e_1 \Rightarrow \sigma_1 \quad \Psi, x : \forall \bar{a}. \sigma_1 \mid e_2 \Rightarrow \sigma_2 \quad \bar{a} \notin \text{ftv}(\Psi)}{\Psi \vdash \text{let } x = e_1 \text{ in } e_2 \Rightarrow \sigma_2}$	$\frac{\text{T-DATA} \quad \Psi, T, D_i : \overline{\sigma_j^j} \rightarrow T \vdash e \Rightarrow \sigma \quad T \notin \text{fT}(\sigma) \quad \Psi \vdash \overline{\sigma_j^j}^i}{\Psi \vdash \text{data } T = \overline{D_i \overline{\sigma_j^j}^i} \text{ in } e \Rightarrow \sigma}$		
$\Psi \vdash e \Leftarrow \sigma$	(Type Checking)		
$\frac{\text{T-LAMC} \quad \Psi, x : \sigma_1 \vdash e \Leftarrow \sigma_2}{\Psi \vdash \lambda x. e \Leftarrow \sigma_1 \rightarrow \sigma_2}$	$\frac{\text{T-FORALL} \quad \Psi \vdash e \Leftarrow \sigma \quad a \notin \text{ftv}(\Psi)}{\Psi \vdash e \Leftarrow \forall a. \sigma}$	$\frac{\text{T-SUB} \quad \Psi \vdash e \Rightarrow \sigma_1 \quad \vdash \sigma_1 <: \sigma_2}{\Psi \vdash e \Leftarrow \sigma_2}$	
$\sigma \triangleright \sigma_1 \rightarrow \sigma_2$	(Matching)		
$\frac{\text{M-FORALL} \quad \sigma[a := \tau] \triangleright \sigma_1 \rightarrow \sigma_2}{\forall a. \sigma \triangleright \sigma_1 \rightarrow \sigma_2}$	$\frac{\text{M-FUNC}}{\sigma_1 \rightarrow \sigma_2 \triangleright \sigma_1 \rightarrow \sigma_2}$		
$\vdash \sigma_1 <: \sigma_2$	(Subtyping)		
$\frac{\text{S-REFL}}{\vdash \sigma <: \sigma}$	$\frac{\text{S-FUNC} \quad \vdash \sigma_3 <: \sigma_1 \quad \vdash \sigma_2 <: \sigma_4}{\vdash \sigma_1 \rightarrow \sigma_2 <: \sigma_3 \rightarrow \sigma_4}$	$\frac{\text{S-FORALLR} \quad \vdash \sigma_1 <: \sigma_2 \quad a \notin \text{ftv}(\sigma_1)}{\vdash \sigma_1 <: \forall a. \sigma_2}$	$\frac{\text{S-FORALLL} \quad \vdash \sigma_1[a := \tau] <: \sigma_2}{\vdash \forall a. \sigma_1 <: \sigma_2}$

Fig. 2. Declarative type system (selected rules)

where rule **M-FORALL** instantiates the type variable a with a monotype. Once rule **T-APP** matches σ to the function type $\sigma_1 \rightarrow \sigma_2$, it checks the argument e_2 against the expected argument type σ_1 , and returns the result type σ_2 . Rule **T-ANNO** ensures the provided annotation is well-formed under the current typing context ($\Psi \vdash \sigma$) to exclude out-of-scope uses of type constructors and checks e against the provided annotation. Rule **T-LET** begins by inferring the type of e_1 , getting σ_1 . It then generalizes σ_1 over variables $\bar{a}$, provided that $\bar{a} \notin \text{ftv}(\Psi)$. The rule then adds $x : \forall \bar{a}. \sigma_1$ to the context to type-check the let body.

Rule **T-DATA** introduces the type constructor and its associated data constructors into the context. It then type-checks e , obtaining the result type σ . Finally, the rule ensures that T does not escape its scope by checking $T \notin \text{fT}(\sigma)$, where fT collects all free type constructors in σ .

Checking. For type-checking, rule **T-LAMC** checks a lambda against a function type $\sigma_1 \rightarrow \sigma_2$, by adding $x : \sigma_1$ to the context and then checking the lambda body against σ_2 . This rule shows

the benefit of bidirectional typing, as it allows the variable x to have a potentially polymorphic type σ_1 . To type-check against a polymorphic type, rule **T-FORALL** first ensures that $a \notin \Psi$, and then proceeds to checking the expression against σ . Lastly, rule **T-SUB** switches from checking to inference mode. It first infers the type of e , and then checks the subtyping relation $\vdash \sigma_1 <: \sigma_2$.

Subtyping. The bottom of Fig. 2 presents the subtyping judgment. Rule **S-REFL** states that a type is a subtype of itself. Rule **S-FUNC** handles function subtyping, where subtyping is contravariant on the argument type, and covariant on the return type. Rule **S-FORALLR** states that σ_1 is a subtype of $\forall a. \sigma_2$, if σ_1 is a subtype of σ_2 , provided that a does not appear free in σ_1 . Lastly, rule **S-FORALLL** instantiates a polymorphic type on the left hand side with a monotone type τ , and checks if $\sigma_1[a := \tau]$ is a subtype of σ_2 .

4 Level-Based Declarative Type System

This section introduces our level-based declarative type system. The language has the same syntax given in Fig. 1. Following Rémy [1992], we assume a given mapping that maps type variables to their levels, which also tracks the levels of type constructors. This mapping implicitly requires that all occurrences of a given type variable must be at the same level. In our examples (§4.2), we always explicitly indicate the level at which each type variable is introduced. (See §5 for a formalism with an explicit level context.) We assume there are infinitely many variables of every level. We write a^n or T^n to denote that a or T is of level n . We can then extend levels to types, where the level of a type is the maximum level of its variables and type constructors. Therefore, types without free type variables or type constructors are always at level zero. We write $\sigma \leq^n$ to denote the type σ with the constraint that its level is at most n .

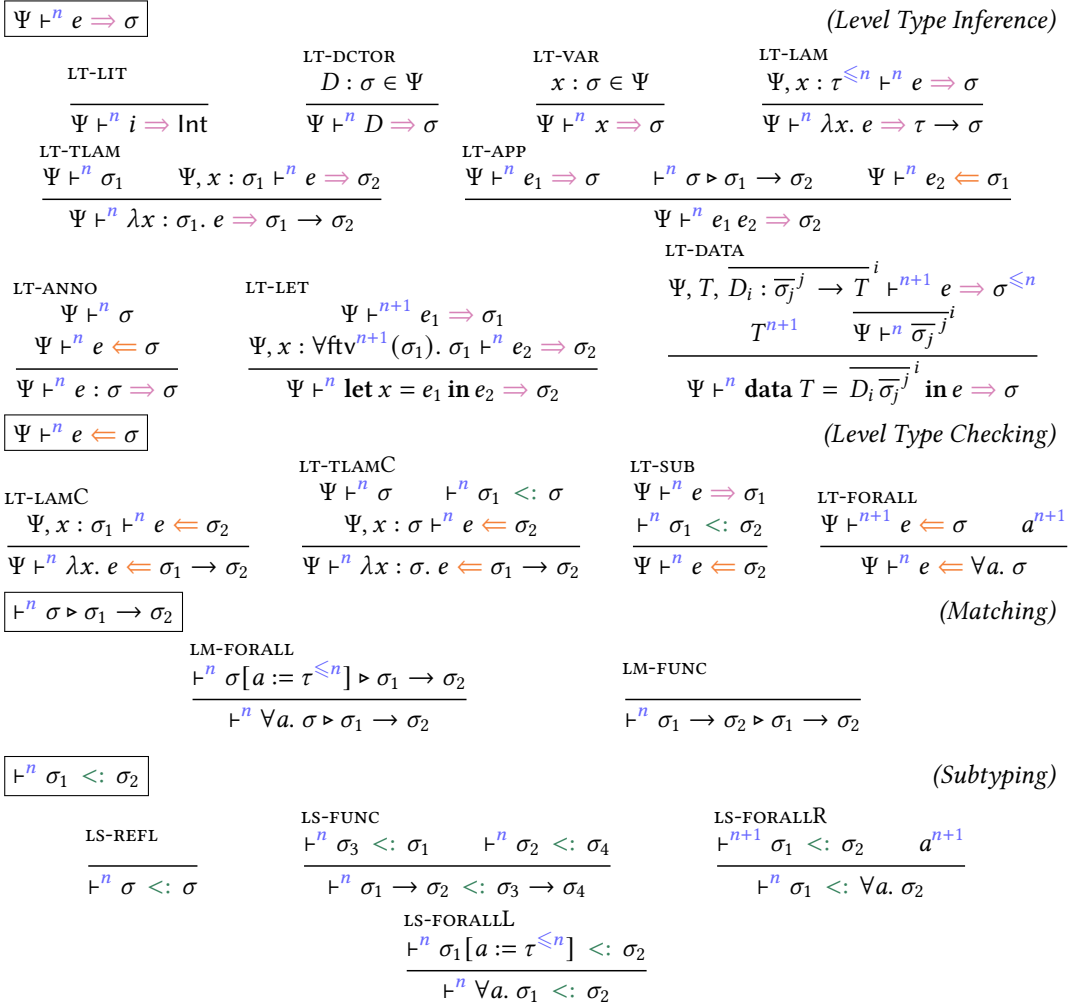
4.1 Typing

Fig. 3 presents the level-based typing rules, where both typing and subtyping are indexed by an integer level n . Rule **LT-LIT** and rule **LT-VAR** are straightforward.

Rule **LT-LAM** again non-deterministically guesses a type τ for the variable x , with the important constraint that τ can be at most level n , the current typing level. Rule **LT-TLAM**, rule **LT-APP**, and rule **LT-ANNO** are self-explanatory. Notably, the matching judgment $\triangleright$ is now also associated with a level. Rule **LT-APP** passes the current typing level to matching, and rule **LM-FORALL** instantiates the polymorphic type with a type at most at level n .

Importantly, rule **LT-LET** increments the level to $n + 1$ when typing the expression e_1 . As a result, lambdas within e_1 can now guess a type at level $n + 1$. After rule **LT-LET** finishes typing e_1 , obtaining type σ_1 , it generalizes all free type variables at $n + 1$ in σ_1 , and adds $x : \forall \text{ftv}^{n+1}(\sigma_1). \sigma_1$ to the context to type-check e_2 at level n . Compared to the previous rule **T-LET** for typing let expressions, this rule does not require traversing the typing context. Rule **LT-DATA** type-checks a local datatype declaration, where the level of T is $n + 1$. The rule adds the type constructor and the associated data constructors to the context, and type-checks e at level $n + 1$, obtaining the result type $\sigma \leq^n$. Since σ is at most level n , it cannot contain T , ensuring that T does not escape from its scope.

Checking. Rule **LT-LAMC** is straightforward. Rule **LT-TLAMC** checks that the expected argument type is a subtype of the parameter type, and adds $x : \sigma$ to the context to check the body. Note that the subtyping relation also takes the current typing level. Rule **LT-SUB** checks that the inferred type is a subtype of the checked type under the current typing level. Rule **LT-FORALL** checks the expression against a polymorphic type. Here, we take a type variable at level $n + 1$, and increment the typing level. Since the type variable is at level $n + 1$ and we ensure that Ψ only contains types at level at most n , this guarantees that existing types in Ψ cannot refer to a , without requiring traversing the typing context.



Generalization. First, consider $\text{let } f = \lambda x. x \text{ in } f$, whose typing derivation is given as follows.

$$\frac{\frac{\bullet, x : a \vdash^1 x \Rightarrow a \quad a^1}{\bullet \vdash^1 \lambda x. x \Rightarrow a \rightarrow a} \text{LT-LAM} \quad \frac{}{f : \forall a. a \rightarrow a \vdash^0 f \Rightarrow \forall a. a \rightarrow a} \text{LT-VAR}}{\bullet \vdash^0 \text{let } f = \lambda x. x \text{ in } f \Rightarrow \forall a. a \rightarrow a} \text{LT-LET}$$

Note that when typing $\lambda x. x$, we are at level 1. We assume a^1 as a side condition, and we can assign $x : a$, since rule **LT-LAM** requires $x : a^{\leq 1}$. As a result, $\text{ftv}^1(a \rightarrow a) = a$, and f gets type $\forall a. a \rightarrow a$.

Notably, using type variables from different levels in the typing derivation can yield different types of the same expression. Specifically, consider the following derivation:

$$\frac{\frac{\bullet, x : b \vdash^1 x \Rightarrow b \quad b^0}{\bullet \vdash^1 \lambda x. x \Rightarrow b \rightarrow b} \text{LT-LAM} \quad \frac{}{f : b \rightarrow b \vdash^0 f \Rightarrow b \rightarrow b} \text{LT-VAR}}{\bullet \vdash^0 \text{let } f = \lambda x. x \text{ in } f \Rightarrow b \rightarrow b} \text{LT-LET}$$

Here we use b^0 as the type of x , and thus the type of f is not generalized. The same type can be derived in the non-level-based system, since rule **T-LET** may not generalize all free type variables in σ_1 .

Subtyping. As another example to demonstrate how the typing of a let binding and subtyping could both increment the level, consider typing $(\text{let } x = (\lambda f : \sigma_1. f \ 2) \ g \text{ in } x)$ at level 0, where

$$\sigma_1 = (\forall a \ b. a \rightarrow b \rightarrow b) \quad \sigma_2 = (\forall c. c \rightarrow \forall d. d \rightarrow d) \quad \Psi = g : \sigma_2$$

Because of the let binding, we need to type $(\lambda f : \sigma_1. f \ 2) \ g$ at level 1, which passes g of type σ_2 to a function expecting an argument of type σ_1 , thus requiring $\vdash^1 \sigma_2 <: \sigma_1$ to hold. We then generalize the let binding by collecting free type variables at level 1. Below we first give the derivation of the subtyping judgment, where some intermediate subderivations are omitted:

$$\textcircled{A} \frac{\frac{\vdash^3 a \rightarrow \forall d. d \rightarrow d <: a \rightarrow b \rightarrow b}{\vdash^3 \forall c. c \rightarrow \forall d. d \rightarrow d <: a \rightarrow b \rightarrow b} \text{LS-FORALLL} \quad a^2 \quad b^3}{\vdash^1 \sigma_2 <: \sigma_1} \text{LS-FORALLR}$$

Note that $\vdash^1 \sigma_2 <: \sigma_1$ holds, as we first skolemize σ_1 with a^2 and b^3 . Then, $\vdash^3 \sigma_2 <: a \rightarrow b \rightarrow b$ holds, as subtyping is now at level 3, and we can instantiate c with a^2 and d with b^3 respectively. On the other hand, $\vdash^1 \sigma_1 <: \sigma_2$ does not hold, as shown in the derivation on the right. At the top of the derivation, we would need to instantiate b with a type at most at level 2. However, when d is skolemized later, it will get a level 3.

$$\frac{\text{not hold} \quad \frac{\vdash^2 \forall b. c \rightarrow b \rightarrow b <: c \rightarrow \forall d. d \rightarrow d}{\vdash^2 \sigma_1 <: c \rightarrow \forall d. d \rightarrow d} \text{LS-FORALLL}}{\vdash^1 \sigma_1 <: \sigma_2} \text{LS-FORALLR}$$

We now type the let binding $(\lambda f : \sigma_1. f \ 2) \ g$, again with some subderivations omitted:

$$\textcircled{B} \frac{\frac{\Psi, f : \sigma_1 \vdash^1 f \Rightarrow \sigma_1 \quad \vdash^1 \sigma_1 \triangleright \text{Int} \rightarrow b_1 \rightarrow b_1 \quad b_1^1}{\Psi, f : \sigma_1 \vdash^1 f \ 2 \Rightarrow b_1 \rightarrow b_1} \text{LT-APP} \quad \frac{\Psi \vdash^1 g \Rightarrow \sigma_2 \quad \textcircled{A}}{\Psi \vdash^1 g \Leftarrow \sigma_1} \text{LT-SUB}}{\Psi \vdash^1 (\lambda f : \sigma_1. f \ 2) \Rightarrow \sigma_1 \rightarrow b_1 \rightarrow b_1} \text{LT-TLAM} \quad \frac{}{\Psi \vdash^1 (\lambda f : \sigma_1. f \ 2) \ g \Rightarrow b_1 \rightarrow b_1} \text{LT-APP}$$

Here we instantiate σ_1 with b_1^1 when typing $f \ 2$. The argument g has type σ_2 , which is a subtype of σ_1 according to $\textcircled{A}$. Note that the return type is $(b_1 \rightarrow b_1)^{\leq 1}$, even though the subtyping derivation $\textcircled{A}$ used a^3 and b^3 . Since both a^2 and b^3 are not used in Ψ , generalization of $b_1 \rightarrow b_1$ only needs to consider free type variables at level 1, but not variables of higher levels, corresponding to rule **LT-LET**.

Putting all pieces together, we have:

$$\frac{(\textcircled{\text{B}}) \quad \Psi, x : \forall b_1. b_1 \rightarrow b_1 \vdash^0 x \Rightarrow \forall b_1. b_1 \rightarrow b_1}{\Psi \vdash^0 \text{let } x = (\lambda f : \sigma_1. f \ 2) \ g \text{ in } x \Rightarrow \forall b_1. b_1 \rightarrow b_1} \text{LT-LET}$$

This example illustrates how multiple rules can increment the typing level and how the level of type variables simultaneously prevents skolem escape and enables let generalization.

5 Coq Mechanization

In this section, we establish the mechanized soundness and completeness results of the level-based declarative type system with respect to the non-level-based system. We begin by outlining the Coq mechanization of the type system, which explicitly encodes level contexts to reason about level-related properties. We then present soundness and completeness.

5.1 Coq Representation

We have assumed an implicit mapping that maps type variables to their levels, which also tracks the levels of type constructors. To facilitate mechanization, we now make level contexts explicit:

$$\text{level context} \quad \Delta ::= \bullet \mid \Delta, a^n \mid \Delta, T^n$$

Level contexts Δ track the levels of both type variables (a^n) and type constructors (T^n). As before, we extend levels to types and contexts, writing $\Delta \vdash \sigma : n$ to denote that the level of σ is n , and $\Delta \vdash \Psi : n$ to denote that the level of Ψ is n .

The typing judgments now incorporate a level context Δ , taking the form $\Delta; \Psi \vdash^n e \Rightarrow \sigma$ and $\Delta; \Psi \vdash^n e \Leftarrow \sigma$. Judgments including matching and subtyping are similarly extended with the level context Δ . These rules are largely unchanged, except for the explicit level handling. For example, rule **LCT-FORALL** adds a^{n+1} into the context. We assume distinct variables in Δ , which is enforced in our mechanization with the locally nameless representation [Charguéraud 2012]. Rule **LCT-LET** uses $\text{ftv}_\Delta^{n+1}(\sigma)$ to generalize $n + 1$ level variables within σ according to the level information in Δ .

$$\frac{\text{LCT-FORALL} \quad \Delta, a^{n+1}; \Psi \vdash^{n+1} e \Leftarrow \sigma}{\Delta; \Psi \vdash^n e \Leftarrow \forall a. \sigma} \quad \frac{\text{LCT-LET} \quad \Delta; \Psi \vdash^{n+1} e_1 \Rightarrow \sigma_1 \quad \Delta; \Psi, x : \forall \text{ftv}_\Delta^{n+1}(\sigma_1). \sigma_1 \vdash^n e_2 \Rightarrow \sigma_2}{\Delta; \Psi \vdash^n \text{let } x = e_1 \text{ in } e_2 \Rightarrow \sigma_2}$$

The level context allows us to reason explicitly about level-related properties. As an example, we prove that at typing level n , if all type variables and type constructors appearing in type context Ψ have a level no greater than the current typing level ($\Delta \vdash \Psi \leq n$), then the inferred type also has a level of at most n ($\Delta \vdash \sigma \leq n$):

Lemma 5.1 (Level of inference mode). *If $\Delta; \Psi \vdash^n e \Rightarrow \sigma$ and $\Delta \vdash \Psi \leq n$, then $\Delta \vdash \sigma \leq n$.*

In typing, $\Delta \vdash \Psi \leq n$ will be maintained as an invariant; at the top level, typing starts with level 0 and an empty context, and $\Delta \vdash \bullet \leq 0$ holds trivially. This justifies that in rule **LT-LET**, generalization only happens at level $n + 1$ by using ftv_Δ^{n+1} .

By explicitly reasoning about these invariants, we can now formally establish the equivalence between the level-based and non-level-based versions of our type system.

5.2 Soundness and Completeness

Soundness. The soundness theorem follows directly. Notably, by maintaining the invariant that all type variables and constructors have levels no greater than the current typing level, introducing a type variable (e.g. rule **LCT-FORALL**) or type constructor at level $n + 1$ ensures freshness relative to the current context. This allows us to establish soundness:

Theorem 5.2 (Soundness of level typing). *Given $\Delta \vdash \Psi \leq n$,*

(Inference) if $\Delta; \Psi \vdash^n e \Rightarrow \sigma$, then $\Psi \vdash e \Rightarrow \sigma$.

(Checking) if $\Delta; \Psi \vdash^n e \Leftarrow \sigma$ where $\Delta \vdash \sigma \leq n$, then $\Psi \vdash e \Leftarrow \sigma$.

In other words, if an expression e has type σ under level n in level-based declarative typing system, then e also has type σ in the non-level-based declarative type system.

Level-related properties. Proving completeness is more subtle, as it requires us to show that for any non-level-based typing $\Psi \vdash e \Leftarrow \sigma$, there *exists* a level context Δ such that $\Delta; \Psi \vdash^n e \Leftarrow \sigma$. However, constructing such a Δ presents a few challenges. First, when typing an application $e_1 e_2$, we need to provide the same level context Δ to derive $\Delta; \Psi \vdash^n e_1 \Rightarrow \sigma$ and $\Delta; \Psi \vdash^n e_2 \Leftarrow \sigma_1$. However, the induction hypothesis provides two distinct level contexts, Δ_1 and Δ_2 , for $\Delta_1; \Psi \vdash^n e_1 \Rightarrow \sigma$ and $\Delta_2; \Psi \vdash^n e_2 \Leftarrow \sigma_1$ respectively. Moreover, rule **LCT-LET** and rule **LCT-FORALL** require finding a Δ such that type context Ψ has a level no greater than the current typing level. Additionally, for generalization, we must ensure that any fresh variables satisfying $\bar{a} \notin \text{ftv}(\Psi)$ (as in rule **T-LET**) must be assigned level $n + 1$ in Δ , while other variables should not.

To address these challenges, we introduce auxiliary definitions that help with merging two level contexts. Specifically,

Definition 5.3 (Level compatibility). *We say that Δ_1 and Δ_2 are compatible at level n , defined as*

$$\Delta_1 \circ^n \Delta_2 \triangleq \forall a^{n_1} (\text{or } T^{n_1}) \in \Delta_1, n_1 \leq n \implies \exists n_2, n_2 \leq n \wedge a^{n_2} (\text{or } T^{n_2}) \in \Delta_2.$$

Definition 5.4 (Level matching). *We say that Δ_1 and Δ_2 match at level n , defined as*

$$\Delta_1 \otimes^n \Delta_2 \triangleq \forall a^{n_1} (\text{or } T^{n_1}) \in \Delta_1, n_1 > n \implies a^{n_1} (\text{or } T^{n_1}) \in \Delta_2.$$

Intuitively, these definitions capture the observations that levels of type variables and constructors can be adjusted with respect to a typing level n . Specifically, compatibility states if a type variable or constructor in Δ_1 has a level no greater than n , its level in Δ_2 remains no greater than n , though the exact levels may differ. Level matching enforces that a type variable or constructor with a level above n in Δ_1 retains exactly the same level in Δ_2 .

Combining these two definitions, we can define level consistency between two level contexts:

Definition 5.5 (Consistency). $\Delta_1 \otimes^n \Delta_2 \triangleq \Delta_1 \circ^n \Delta_2 \wedge \Delta_1 \otimes^n \Delta_2$.

Lemma 5.6 (Consistency preserves typing). *Given $\Delta \vdash \Psi \leq n$, and $\Delta_1 \otimes^n \Delta_2$, (1) if $\Delta_1; \Psi \vdash^n e \Rightarrow \sigma$, then $\Delta_2; \Psi \vdash^n e \Rightarrow \sigma$; and (2) if $\Delta_1; \Psi \vdash^n e \Leftarrow \sigma$ where $\Delta \vdash \sigma \leq n$, then $\Delta_2; \Psi \vdash^n e \Leftarrow \sigma$.*

With these definitions and properties, we can now rename variables and adjust their levels in the level contexts when needed to resolve the challenges.

Completeness. We prove completeness:

Theorem 5.7 (Completeness of level typing).

(Inference) If $\Psi \vdash e \Rightarrow \sigma$, then there exists Δ and n , such that $\Delta \vdash \Psi \leq n$ and $\Delta; \Psi \vdash^n e \Rightarrow \sigma$.

(Checking) If $\Psi \vdash e \Leftarrow \sigma$, then there exist Δ and n , such that $\Delta \vdash \Psi \leq n$ and $\Delta \vdash \sigma \leq n$ and $\Delta; \Psi \vdash^n e \Leftarrow \sigma$.

With that, we establish the equivalence between the level-based and non-level-based versions of the type system.

polytype	σ	$::=$	$\forall a. \sigma \mid \sigma_1 \rightarrow \sigma_2 \mid \tau$
monotype	τ	$::=$	$\text{Int} \mid a \mid T \mid \tau_1 \rightarrow \tau_2 \mid \hat{a}$
term context	Σ	$::=$	$\bullet \mid \Sigma, x : \sigma \mid \Sigma, D : \sigma$
algorithmic context	Γ, Θ, Δ	$::=$	$\bullet \mid \Gamma, T^n \mid \Gamma, a^n \mid \Gamma, \hat{a}^n \mid \Gamma, \hat{a}^n = \tau$
complete context	Ω	$::=$	$\bullet \mid \Gamma, T^n \mid \Gamma, a^n \mid \Gamma, \hat{a}^n = \tau$

$[\Gamma]\sigma$

(Context Application)

$[\Gamma]\text{Int} = \text{Int}$	$[\Gamma](\sigma_1 \rightarrow \sigma_2) = [\Gamma]\sigma_1 \rightarrow [\Gamma]\sigma_2$	$[\Gamma]\hat{a} = \hat{a} \quad \text{if } \hat{a} \notin \Gamma \text{ or } \hat{a}^n \in \Gamma$
$[\Gamma]a = a$	$[\Gamma](\forall a. \sigma) = \forall a. [\Gamma]\sigma$	$[\Gamma]\hat{a} = [\Gamma]\tau \quad \text{if } \hat{a}^n = \tau \in \Gamma$
$[\Gamma]T = T$		

$\Gamma \vdash^n \sigma$

(Type Well-Formedness)

$\Gamma \vdash^n \text{Int}$	$\frac{a^m \in \Gamma}{m \leq n} \quad \frac{T^m \in \Gamma}{m \leq n}$ $\frac{}{\Gamma \vdash^n a} \quad \frac{}{\Gamma \vdash^n T}$	$\frac{\Gamma \vdash^n \sigma_1 \quad \Gamma \vdash^n \sigma_2}{\Gamma \vdash^n \sigma_1 \rightarrow \sigma_2}$	$\frac{\hat{a}^m \in \Gamma}{m \leq n} \quad \frac{\hat{a}^m = \tau \in \Gamma}{m \leq n}$ $\frac{}{\Gamma \vdash^n \hat{a}} \quad \frac{}{\Gamma \vdash^n \hat{a}}$	$\frac{\Gamma, a^n \vdash^n \sigma}{\Gamma \vdash^n \forall a. \sigma}$
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Fig. 4. Syntax of the algorithmic system, context application, and well-formedness of types

6 Algorithmic Type System with Levels

This section first presents the algorithmic type system with levels, and then shows that the algorithmic system is sound and complete with respect to the level-based declarative type system.

Fig. 4 presents the syntax of the algorithmic system. Monomorphic types are extended with unification variables $\hat{a}$, representing unknown types that will be inferred.

We have two typing contexts: a term context Σ that maps local variables ($x : \sigma$) and data constructors ($D : \sigma$) to their types, and an algorithmic context Γ that tracks levels of type constructors (T^n), type variables (a^n), and unification variables ($\hat{a}^n$). Additionally, Γ records the solutions for unification variables ($\hat{a}^n = \tau$), with the invariant that τ has a level no greater than n . A complete context Ω is an algorithmic context in which all unification variables have been solved. Notably, the contexts are not ordered, unlike Dunfield and Krishnaswami [2013]. Since contexts contain solutions for unification variables, we use $[\Gamma]\sigma$ to denote the type obtained by applying Γ as a substitution to σ .

Well-formedness of types $\Gamma \vdash^n \sigma$ denotes that σ is well-formed under the algorithmic context Γ at level n . It checks that all type variables, unification variables, and type constructors are bound in the context, and that their levels are no greater than the typing level.

6.1 Algorithmic Typing

Fig. 5 presents the algorithmic typing rules. The typing judgment $\Gamma \mid \Sigma \vdash^n e \Rightarrow \sigma \vdash \Delta$ (and $\Gamma \mid \Sigma \vdash^n e \Leftarrow \sigma \vdash \Delta$) reads: under the algorithmic context Γ and term context Σ , at typing level n , expression e infers (or checks against) type σ , updating the algorithmic context to Δ . Intuitively, the algorithmic context is threaded through algorithmic judgments and accumulates information.

Rule **AT-LIT**, rule **AT-DCTOR**, and rule **AT-VAR** are straightforward, and all return the algorithmic context unchanged. Rule **AT-LAM**, instead of guessing a monotype for x as in the declarative system, creates a new unification variable $\hat{a}^n$ of the current typing level in the algorithmic context, and adds $x : \hat{a}$ to the context. We assume that new unification variables introduced to the context are always fresh. By assigning $\hat{a}^n$, we constrain its solution to a type with a level no greater than n , thus effectively ensuring that x gets a type of a level no greater than n . The rule then proceeds to type-check the lambda body, updating the algorithmic context accordingly. Rule **AT-TLAM** simply

$$\boxed{\Gamma \mid \Sigma \vdash^n e \Rightarrow \sigma \vdash \Delta} \quad \left(\begin{array}{l} \text{Algorithmic Level Type Inference} \\ \text{Inputs: } \Gamma, \Sigma, n, e; \text{ Outputs: } \sigma, \Delta \end{array} \right)$$

$$\begin{array}{c}
\text{AT-LIT} \quad \frac{}{\Gamma \mid \Sigma \vdash^n i \Rightarrow \text{Int} \vdash \Gamma} \quad \text{AT-DCTOR} \quad \frac{D : \sigma \in \Sigma}{\Gamma \mid \Sigma \vdash^n D \Rightarrow \sigma \vdash \Gamma} \quad \text{AT-VAR} \quad \frac{x : \sigma \in \Sigma}{\Gamma \mid \Sigma \vdash^n x \Rightarrow \sigma \vdash \Gamma} \\
\text{AT-LAM} \quad \frac{\Gamma, \hat{\alpha}^n \mid \Sigma, x : \hat{\alpha}^n e \Rightarrow \sigma \vdash \Delta}{\Gamma \mid \Sigma \vdash^n \lambda x. e \Rightarrow \hat{\alpha} \rightarrow \sigma \vdash \Delta} \quad \text{AT-TLAM} \quad \frac{\Gamma \vdash^n \sigma_1 \quad \Gamma \mid \Sigma, x : \sigma_1 \vdash^n e \Rightarrow \sigma_2 \vdash \Delta}{\Gamma \mid \Sigma \vdash^n \lambda x : \sigma_1. e \Rightarrow \sigma_1 \rightarrow \sigma_2 \vdash \Delta} \quad \text{AT-ANNO} \quad \frac{\Gamma \vdash^n \sigma \quad \Gamma \mid \Sigma \vdash^n e \Leftarrow \sigma \vdash \Delta}{\Gamma \mid \Sigma \vdash^n e : \sigma \Rightarrow \sigma \vdash \Delta} \\
\text{AT-APP} \quad \frac{\Gamma \mid \Sigma \vdash^n e_1 \Rightarrow \sigma \vdash \Theta_1 \quad \Theta_1 \vdash^n [\Theta_1] \sigma \triangleright \sigma_1 \rightarrow \sigma_2 \vdash \Theta_2 \quad \Theta_2 \mid \Sigma \vdash^n e_2 \Leftarrow [\Theta_2] \sigma_1 \vdash \Delta}{\Gamma \mid \Sigma \vdash^n e_1 e_2 \Rightarrow \sigma_2 \vdash \Delta} \\
\text{AT-LET} \quad \frac{\Gamma \mid \Sigma \vdash^{n+1} e_1 \Rightarrow \sigma_1 \vdash \Theta \quad \text{fuv}_{\Theta}^{n+1}([\Theta] \sigma_1) = \bar{\alpha} \quad \Theta \mid \Sigma, x : \forall \bar{a}. (([\Theta] \sigma_1)[\bar{\alpha} := \bar{a}]) \vdash^n e_2 \Rightarrow \sigma_2 \vdash \Delta}{\Gamma \mid \Sigma \vdash^n \text{let } x = e_1 \text{ in } e_2 \Rightarrow \sigma_2 \vdash \Delta} \\
\text{AT-DATA} \quad \frac{\Gamma \vdash^n \bar{\sigma}_j^j \quad \Gamma, T^{n+1} \mid \Sigma, D_i : \bar{\sigma}_j^j \rightarrow T \vdash^{n+1} e \Rightarrow \sigma \vdash \Delta \quad \Delta \vdash^n \sigma}{\Gamma \mid \Sigma \vdash^n \text{data } T = \bar{D}_i \bar{\sigma}_j^j \text{ in } e \Rightarrow \sigma \vdash \Delta_T}
\end{array}$$

$$\boxed{\Gamma \mid \Sigma \vdash^n e \Leftarrow \sigma \vdash \Delta} \quad \left(\begin{array}{l} \text{Algorithmic Level Type Checking} \\ \text{Inputs: } \Gamma, \Sigma, n, e, \sigma; \text{ Output: } \Delta \end{array} \right)$$

$$\begin{array}{c}
\text{AT-LAMC} \quad \frac{\Gamma \mid \Sigma, x : \sigma_1 \vdash^n e \Leftarrow \sigma_2 \vdash \Delta}{\Gamma \mid \Sigma \vdash^n \lambda x. e \Leftarrow \sigma_1 \rightarrow \sigma_2 \vdash \Delta} \quad \text{AT-TLAMC} \quad \frac{\Gamma \vdash^n \sigma_1 <: \sigma \vdash \Theta \quad \Theta \mid \Sigma, x : \sigma \vdash^n e \Leftarrow [\Theta] \sigma_2 \vdash \Delta_2}{\Gamma \mid \Sigma \vdash^n \lambda x : \sigma. e \Leftarrow \sigma_1 \rightarrow \sigma_2 \vdash \Delta_2} \\
\text{AT-SUB} \quad \frac{\Gamma \mid \Sigma \vdash^n e \Rightarrow \sigma_1 \vdash \Theta \quad \Theta \vdash^n [\Theta] \sigma_1 <: [\Theta] \sigma_2 \vdash \Delta}{\Gamma \mid \Sigma \vdash^n e \Leftarrow \sigma_2 \vdash \Delta} \quad \text{AT-FORALL} \quad \frac{\Gamma, a^{n+1} \mid \Sigma \vdash^{n+1} e \Leftarrow \sigma \vdash \Delta}{\Gamma \mid \Sigma \vdash^n e \Leftarrow \forall a. \sigma \vdash \Delta}
\end{array}$$

$$\boxed{\Gamma \vdash^n \sigma \triangleright \sigma_1 \rightarrow \sigma_2 \vdash \Delta} \quad \left(\begin{array}{l} \text{Algorithmic Matching} \\ \text{Inputs: } \Gamma, n, \sigma; \text{ Outputs: } \sigma_1, \sigma_2, \Delta \end{array} \right)$$

$$\begin{array}{c}
\text{AM-FORALL} \quad \frac{\Gamma, \hat{\alpha}^n \vdash^n \sigma[a := \hat{\alpha}] \triangleright \sigma_1 \rightarrow \sigma_2 \vdash \Delta}{\Gamma \vdash^n \forall a. \sigma \triangleright \sigma_1 \rightarrow \sigma_2 \vdash \Delta} \quad \text{AM-FUNC} \quad \frac{}{\Gamma \vdash^n \sigma_1 \rightarrow \sigma_2 \triangleright \sigma_1 \rightarrow \sigma_2 \vdash \Gamma} \\
\text{AM-UVAR} \quad \frac{}{\Gamma, \hat{\alpha}^m, \Gamma' \vdash^n \hat{\alpha} \triangleright \hat{\alpha}_1 \rightarrow \hat{\alpha}_2 \vdash \Gamma, \hat{\alpha}^m = \hat{\alpha}_1 \rightarrow \hat{\alpha}_2, \Gamma', \hat{\alpha}_1^m, \hat{\alpha}_2^m}
\end{array}$$

Fig. 5. Algorithmic typing

adds $x : \sigma_1$ to the context to type-check the body. Rule **AT-ANNO** checks the expression against the provided type annotation.

Rule **AT-APP** first infers the type of e_1 , obtaining σ and updating the algorithmic context to Θ_1 . Next, the rule applies the matching judgment, given at the bottom of Fig. 5, to instantiate $[\Theta_1] \sigma$ to a function type. Compared to the declarative system, the algorithmic matching judgment additionally takes both the typing level and the algorithmic context. There are three rules. Rule **AM-FORALL** instantiates a polymorphic type with a new unification variable of the given level n , and matches

$\boxed{\Gamma \vdash^n \sigma_1 <: \sigma_2 \vdash \Delta}$ <i>(Algorithmic Subtyping Inputs: $\Gamma, n, \sigma_1, \sigma_2$; Output: Δ)</i>			
AS-VAR	AS-INT	AS-TYCTOR	AS-UVAR
$\frac{}{\Gamma \vdash^n a <: a \vdash \Gamma}$	$\frac{}{\Gamma \vdash^n \text{Int} <: \text{Int} \vdash \Gamma}$	$\frac{}{\Gamma \vdash^n T <: T \vdash \Gamma}$	$\frac{}{\Gamma \vdash^n \hat{a} <: \hat{a} \vdash \Gamma}$
AS-FUNC			
$\frac{\Gamma \vdash^n \sigma_3 <: \sigma_1 \vdash \Theta \quad \Theta \vdash^n [\Theta] \sigma_2 <: [\Theta] \sigma_4 \vdash \Delta}{\Gamma \vdash^n \sigma_1 \rightarrow \sigma_2 <: \sigma_3 \rightarrow \sigma_4 \vdash \Delta}$	AS-FORALLR $\frac{\Gamma, a^{n+1} \vdash^{n+1} \sigma_1 <: \sigma_2 \vdash \Delta}{\Gamma \vdash^n \sigma_1 <: \forall a. \sigma_2 \vdash \Delta}$	AS-FORALLL $\frac{\Gamma, \hat{a}^n \vdash^n \sigma_1 [a := \hat{a}] <: \sigma_2 \vdash \Delta}{\Gamma \vdash^n \forall a. \sigma_1 <: \sigma_2 \vdash \Delta}$	
AS-SOLVEL $\frac{\hat{a} \notin \text{fuv}(\sigma) \quad \Gamma, \hat{a}^m, \Gamma' \vdash \sigma \rightsquigarrow_{\bar{m}} \tau \vdash \Delta, \hat{a}^m, \Delta'}{\Gamma, \hat{a}^m, \Gamma' \vdash^n \hat{a} <: \sigma \vdash \Delta, \hat{a}^m = \tau, \Delta'}$		AS-SOLVER $\frac{\hat{a} \notin \text{fuv}(\sigma) \quad \Gamma, \hat{a}^m, \Gamma' \vdash \sigma \rightsquigarrow_{\bar{m}}^+ \tau \vdash \Delta, \hat{a}^m, \Delta'}{\Gamma, \hat{a}^m, \Gamma' \vdash^n \sigma <: \hat{a} \vdash \Delta, \hat{a}^m = \tau, \Delta'}$	

Fig. 6. Algorithmic subtyping

$\boxed{\Gamma \vdash \sigma \rightsquigarrow_{\bar{m}}^{\pm} \tau \vdash \Delta}$ $\boxed{\text{polarity } \pm ::= + \mid -}$			<i>(Polymorphic Promotion Inputs: $\Gamma, \sigma, \pm, m$; Outputs: τ, Δ)</i>
PR-INT	PR-SK	PR-TYCTOR	
$\frac{}{\Gamma \vdash \text{Int} \rightsquigarrow_{\bar{m}}^{\pm} \text{Int} \vdash \Gamma}$	$\frac{a^{m_1} \in \Gamma \quad m_1 \leq m_2}{\Gamma \vdash a \rightsquigarrow_{\bar{m}_2}^{\pm} a \vdash \Gamma}$	$\frac{T^{m_1} \in \Gamma \quad m_1 \leq m_2}{\Gamma \vdash T \rightsquigarrow_{\bar{m}_2}^{\pm} T \vdash \Gamma}$	
PR-UVAR	PR-UVARPR		
$\frac{\hat{a}^{m_1} \in \Gamma \quad m_1 \leq m_2}{\Gamma \vdash \hat{a} \rightsquigarrow_{\bar{m}_2}^{\pm} \hat{a} \vdash \Gamma}$	$\frac{m_1 > m_2}{\Gamma, \hat{a}_1^{m_1}, \Gamma' \vdash \hat{a}_1 \rightsquigarrow_{\bar{m}_2}^{\pm} \hat{a}_2 \vdash \Gamma, \hat{a}_1^{m_1} = \hat{a}_2, \Gamma', \hat{a}_2^{m_2}}$		
PR-FUNC			
$\frac{\Gamma \vdash \sigma_1 \rightsquigarrow_{\bar{m}}^{\mp} \tau_1 \vdash \Theta \quad \Theta \vdash [\Theta] \sigma_2 \rightsquigarrow_{\bar{m}}^{\pm} \tau_2 \vdash \Delta}{\Gamma \vdash \sigma_1 \rightarrow \sigma_2 \rightsquigarrow_{\bar{m}}^{\pm} \tau_1 \rightarrow \tau_2 \vdash \Delta}$	PR-FORALLPOS $\frac{\Gamma, \hat{a}^m \vdash \sigma [a := \hat{a}] \rightsquigarrow_{\bar{m}}^+ \tau \vdash \Delta}{\Gamma \vdash \forall a. \sigma \rightsquigarrow_{\bar{m}}^+ \tau \vdash \Delta}$	PR-FORALLNEG $\frac{\Gamma, a^{m+1} \vdash \sigma \rightsquigarrow_{\bar{m}}^- \tau \vdash \Delta}{\Gamma \vdash \forall a. \sigma \rightsquigarrow_{\bar{m}}^- \tau \vdash \Delta}$	

Fig. 7. Polymorphic promotion

the body. Rule **AM-FUNC** directly returns the input function. Lastly, rule **AM-UVAR** handles unification variables. In this case, the variable must be unsolved, at some level m . Since the variable's solution must be a function, we create two new unification variables $\hat{a}_1^m$ and $\hat{a}_2^m$, both at the level as m , and set $\hat{a}^m = \hat{a}_1 \rightarrow \hat{a}_2$. Once matching $[\Theta_1]\sigma$ in rule **AT-APP** returns $\sigma_1 \rightarrow \sigma_2$, the rule checks the argument with the expected type $[\Theta_2]\sigma_1$, resulting in the final algorithmic context Δ .

Rule **AT-LET** type-checks let expressions. The rule begins by incrementing the typing level to type-check e_1 , obtaining σ_1 . Then, it collects the unsolved unification variables $\text{fuv}_{\Theta}^{n+1}$ at level $n+1$ within $[\Theta]\sigma_1$ (using the level information in Θ), resulting in a set of unification variables $\bar{a}$. Next, it generalizes these unification variables by substituting them with fresh type variables $\bar{a}$ within $[\Theta]\sigma_1$, obtaining $\forall \bar{a}. (([\Theta]\sigma_1)[\bar{a} := \bar{a}])$. The unification variables $\bar{a}$ will no longer be useful and may be removed from the algorithmic context, although this is not strictly required. Finally, it adds x of the generalized type to the context, and type-checks the let body at level n . Lastly, rule **AT-DATA** adds T^{n+1} to the algorithmic context and associated data constructors to the context to type-check e under $n+1$, obtaining σ . It then checks that σ is well-typed under level n . The returned algorithmic context is $\Delta_{\setminus T}$, which removes all the occurrences of T^{n+1} from the output context and its complete definition can be found in the appendix.

Type checking. For checking, we maintain the invariant that the type used for checking is fully substituted by the current algorithmic context. Rule **AT-LAMC** is self-explanatory. Rule **AT-TLAMC** checks that σ_1 is a subtype of σ , where the subtyping judgment takes the algorithmic context and returns a new Θ . Since Θ may contain new solutions for unification variables, we apply it to σ_2 when checking the lambda body. Rule **AT-SUB** first infers the type of e , obtaining σ_1 and a new Θ . The rule then applies the context to the types for subtyping, and thus the input types to subtyping are also fully substituted. Lastly, rule **AT-FORALL** adds the type variable a^{n+1} to the algorithmic context and increments the typing level to check e .

6.2 Subtyping

Fig. 6 presents the algorithmic subtyping rules. The judgment $\Gamma \vdash^n \sigma_1 <: \sigma_2 \vdash \Delta$ reads: under the algorithmic context Γ and at level n , type σ_1 is a subtype of σ_2 , updating the algorithmic context to Δ . We maintain the invariant that the input types σ_1 and σ_2 are fully substituted under Γ , and thus rules **AS-SOLVE**L and **AS-SOLVE**R only deal with cases with unsolved unification variables.

The first four rules are straightforward. In rule **AS-FUNC**, subtyping is contravariant over function argument types, and covariant over return types. Note that the rule applies the context Θ to σ_2 and σ_4 , as Θ may contain new information about unification variables. Rule **AS-FORALLR** skolemizes the polymorphic type with a new type variable a^{n+1} , and increments the subtyping level. Rule **AS-FORALLL** instantiates the polymorphic type with a new unification variable of the current level.

Of particular interest are the last two rules, which involve unification variables. Rule **AS-SOLVE**L requires $\hat{\alpha}$ to be a subtype of σ , while rule **AS-SOLVE**R requires σ to be a subtype of $\hat{\alpha}$. In both cases, the rule performs occurs-check ($\hat{\alpha} \notin \text{ftv}(\sigma)$). Then, it uses the *promotion* judgment to promote σ to a monotype τ . This process is discussed below. The result monotype τ is guaranteed to be well-typed at the promotion level m . Therefore, we set $\hat{\alpha}^m = \tau$ in the output algorithmic context.⁵

Polymorphic promotion. Fig. 7 presents the novel polymorphic promotion judgment. The judgment $\Gamma \vdash \sigma \rightsquigarrow_m^\pm \tau \vdash \Delta$ reads: under the algorithmic context Γ and at level m , promoting type σ under polarity $\pm$ produces a monotype τ , updating the algorithmic context to Δ . Intuitively, the polarity $\pm$ indicates that the type being promoted is a subtype (+) or supertype (−) of a type variable of level m . Since m indicates the promoted level, it never changes in the rules. Recall that rule **AS-SOLVE**L uses promotion under (−), while rule **AS-SOLVE**R uses it under (+).

Rule **PR-INT** returns the type unchanged. Rule **PR-SK** promotes a type variable a^{m_1} . This rule requires that the variable's level m_1 be no greater than the promotion level m_2 ($m_1 \leq m_2$), as the promotion result will become part of the solution for a unification variable at level m_2 , which cannot refer to type variables of higher levels. In other words, promoting a type variable can (correctly) fail. Such failure corresponds to the case where a type variable would otherwise escape its scope through a unification variable. Similarly, rule **PR-TYCTOR** requires T 's level m_1 be no greater than m_2 .

Promoting unification variables involves two rules: rule **PR-UVAR** and rule **PR-UVARPR**. Intuitively, a unification variable of a greater level can be promoted to a smaller level for it to be part of a solution for a unification variable with a smaller level. Specifically, if the unification variable's level is no greater than the promotion level, rule **PR-UVAR** returns the variable unchanged. Otherwise, rule **PR-UVARPR** adjusts the level by introducing a new unification variable $\hat{\alpha}_2^{m_2}$ at level m_2 , and setting $\hat{\alpha}_1^{m_1} = \hat{\alpha}_2$.

Rule **PR-FUNC** promotes function types. Due to contravariant function typing, the rule first promotes the argument type under the flipped polarity (denoted as $\mp$), obtaining τ_1 and Θ . Then, it promotes the result type $[\Theta]\sigma_2$ to τ_2 under the original polarity. The final promoted type is $\tau_1 \rightarrow \tau_2$.

⁵There is overlapping between, e.g. rule **AS-SOLVE**L and rule **AS-FORALLL**. Such overlap is benign, as their application produces equivalent results. Deterministic behavior could be enforced with additional side conditions.

$$\begin{array}{c}
\Delta_1 = \hat{\alpha}^0, \hat{\beta}^1 \\
\Delta_2 = \hat{\alpha}^0, \hat{\beta}^1 = \hat{\beta}_1, \hat{\beta}_1^0 \\
\Delta_3 = \hat{\alpha}^0 = \hat{\beta}_1 \rightarrow \hat{\beta}_1, \hat{\beta}^1 = \hat{\beta}_1, \hat{\beta}_1^0 \\
\textcircled{1} \frac{\hat{\alpha}^0 \mid \Sigma, x : \hat{\alpha} \vdash^1 f \Rightarrow \sigma \vdash \hat{\alpha}^0}{\hat{\alpha}^0 \vdash^1 \sigma \triangleright (\hat{\beta} \rightarrow \hat{\beta}) \rightarrow \hat{\beta} \vdash \Delta_1} \quad \frac{\Delta_1 \mid \Sigma, x : \hat{\alpha} \vdash^1 x \Rightarrow \hat{\alpha} \vdash \Delta_1 \quad \Delta_1 \vdash^1 \hat{\alpha} <: \hat{\beta} \rightarrow \hat{\beta} \vdash \Delta_3 \textcircled{3}}{\Delta_1 \mid \Sigma, x : \hat{\alpha} \vdash^1 x \Leftarrow \hat{\beta} \rightarrow \hat{\beta} \vdash \Delta_3} \text{AT-SUB} \\
\hline
\hat{\alpha}^0 \mid \Sigma, x : \hat{\alpha} \vdash^1 f x \Rightarrow \hat{\beta} \vdash \Delta_3 \quad \text{AT-APP}
\end{array}$$

PR-UVARPr PR-UVAR AS-FUNC AS-SOLVEL

Fig. 8. Example derivation

Promoting polymorphic types depends on polarity. Rule **PR-FORALLPos** promotes a polymorphic type $\forall a. \sigma$ under (+). This means that the polymorphic type $\forall a. \sigma$ needs to be a subtype of a monotype. Thus, we instantiate a with a fresh unification variable $\hat{\alpha}^m$ of the promotion level m . Conversely, rule **PR-FORALLNeg** promotes a polymorphic type under (-), which requires $\forall a. \sigma$ to be a supertype of a monotype. The rule instantiates a with a fresh type variable at level $m + 1$, while promotion stays at level m , effectively preventing a from appearing in σ .

Examples. To see how polarity works, consider the following derivations, with $\hat{\alpha}^0$, for $\forall b. b \rightarrow b <: \hat{\alpha}$ on the left, and $\hat{\alpha} <: \forall b. b \rightarrow b$ on the right, respectively:

$$\begin{array}{c}
\frac{\hat{\alpha}^0, \hat{\beta}^0 \vdash \hat{\beta} \rightarrow \hat{\beta} \rightsquigarrow_0^+ \hat{\beta} \rightarrow \hat{\beta} \vdash \hat{\alpha}^0, \hat{\beta}^0}{\hat{\alpha}^0 \vdash \forall b. b \rightarrow b \rightsquigarrow_0^+ \hat{\beta} \rightarrow \hat{\beta} \vdash \hat{\alpha}^0, \hat{\beta}^0} \text{PR-FORALLPos} \quad \frac{\hat{\alpha}^0, b^1 \vdash b \rightarrow b \rightsquigarrow_0^- ? \vdash ?}{\hat{\alpha}^0 \vdash \forall b. b \rightarrow b \rightsquigarrow_0^- ? \vdash ?} \text{PR-FORALLNeg} \\
\hline
\hat{\alpha}^0 \vdash^0 \forall b. b \rightarrow b <: \hat{\alpha} \vdash \hat{\alpha}^0 = \hat{\beta} \rightarrow \hat{\beta} \vdash \hat{\beta}^0 \quad \text{AS-SOLVEL} \quad \hat{\alpha}^0 \vdash^0 \hat{\alpha} <: \forall b. b \rightarrow b \vdash ? \quad \text{AS-SOLVEL}
\end{array}$$

In the left case, we promote $\forall b. b \rightarrow b$ under (+), allowing us to instantiate b with $\hat{\beta}^0$. This is valid as $\forall b. b \rightarrow b$ is indeed a subtype of any monotype $\tau \rightarrow \tau$. Conversely, in the right case, we instantiate b with b^1 , and promoting b will fail, since rule **PR-SK** does not apply. This failure is expected, as indeed no monotype can be a subtype of $\forall b. b \rightarrow b$.

We now consider a larger example to see how things work together. Specifically, consider typing $(\lambda x. \text{let } y = f x \text{ in } y)$ under $\Sigma = f : \sigma$, where $\sigma = \forall a. (a \rightarrow a) \rightarrow a$, with the following derivation:

$$\begin{array}{c}
\mathcal{D} \\
\frac{\hat{\alpha}^0 \mid \Sigma, x : \hat{\alpha} \vdash^1 f x \Rightarrow \hat{\beta} \vdash \Delta_3 \quad \text{ftv}_{\Delta_3}^1([\Delta_3]\hat{\beta}) = \emptyset \quad \Delta_3 \mid \Sigma, x : \hat{\alpha}, y : \hat{\beta}_1 \vdash^0 y \Rightarrow \hat{\beta}_1 \vdash \Delta_3}{\hat{\alpha}^0 \mid \Sigma, x : \hat{\alpha} \vdash^0 (\text{let } y = f x \text{ in } y) \Rightarrow \hat{\beta}_1 \vdash \Delta_3} \text{AT-LET} \\
\hline
\bullet \mid \Sigma \vdash^0 (\lambda x. \text{let } y = f x \text{ in } y) \Rightarrow \hat{\alpha} \rightarrow \hat{\beta}_1 \vdash \Delta_3 \quad \text{AT-LAM}
\end{array}$$

Here, rule **AT-LAM** creates a new unification variable $\hat{\alpha}^0$ as the type of $x : \hat{\alpha}$. Rule **AT-LET** type-checks $f x$, and generalizes the result as the type of y . The derivation $\mathcal{D}$ is given in Fig. 8.

There are a few notable things. First, at $\textcircled{1}$, we match f 's type σ to $(\hat{\beta} \rightarrow \hat{\beta}) \rightarrow \hat{\beta}$, with $\hat{\beta}^1$. Then, rule **AT-SUB** checks if x 's type $\hat{\alpha}$ is a subtype of the expected argument type $\hat{\beta} \rightarrow \hat{\beta}$. At $\textcircled{2}$, rule **AS-SOLVEL** applies, promoting $\hat{\beta} \rightarrow \hat{\beta}$ under 0, which is $\hat{\alpha}$'s level. Rule **PR-UVARPr** promotes $\hat{\beta}$ by creating a new unification variable $\hat{\beta}_1^0$, and sets $\hat{\beta}^1 = \hat{\beta}_1$, effectively lowering $\hat{\beta}$'s level to 0. Then, rule **PR-UVAR** promotes $[\Delta_2]\hat{\beta} = \hat{\beta}_1$, returning $\hat{\beta}_1$. Therefore, at $\textcircled{3}$, promotion succeeds, and we set $\hat{\alpha} = \hat{\beta}_1 \rightarrow \hat{\beta}_1$. As the final result, rule **AT-APP** returns $\hat{\beta}$ and the typing context Δ_3 .

Returning to rule **AT-LET**, there are no level 1 variables within $[\Delta_3]\hat{\beta} = \hat{\beta}_1$. Thus y has type $\hat{\beta}_1$, and the final type is $\hat{\alpha} \rightarrow \hat{\beta}_1$.

$$\begin{array}{c}
\boxed{\Gamma \vdash^n \Sigma} \\
\hline
\Gamma \vdash^n \bullet
\end{array}
\quad
\frac{\Gamma \vdash^n \sigma \quad \Gamma \vdash^n \Sigma \quad x \notin \text{dom}(\Sigma)}{\Gamma \vdash^n \Sigma, x : \sigma}
\quad
\frac{\Gamma \vdash^n \sigma \quad \Gamma \vdash^n \Sigma \quad D \notin \text{dom}(\Sigma)}{\Gamma \vdash^n \Sigma, D : \sigma}
\quad
\text{(Term Context Well-Formedness)}$$

$$\begin{array}{c}
\boxed{\Gamma \vdash^n \Delta} \\
\hline
\Gamma \vdash^n \bullet
\end{array}
\quad
\frac{\Gamma \vdash^n \Delta \quad a \notin \Delta \quad m \leq n}{\Gamma \vdash^n \Delta, a^m}
\quad
\frac{\Gamma \vdash^n \Delta \quad T \notin \Delta \quad m \leq n}{\Gamma \vdash^n \Delta, T^m}
\quad
\text{(Algorithmic Context Well-Formedness)}$$

$$\frac{\Gamma \vdash^n \Delta \quad \hat{\alpha} \notin \Delta \quad m \leq n}{\Gamma \vdash^n \Delta, \hat{\alpha}^m}
\quad
\frac{\Gamma \vdash^n \Delta \quad \Gamma \vdash^m \tau \quad \hat{\alpha} \notin \text{fv}([\Delta]\tau) \quad \hat{\alpha} \notin \Delta \quad m \leq n}{\Gamma \vdash^n \Delta, \hat{\alpha}^m = \tau}$$

Fig. 9. Well-formedness of contexts

$$\boxed{\Gamma \longrightarrow \Delta}
\quad
\text{(Context Extension)}$$

$$\begin{array}{c}
\bullet \longrightarrow \bullet \quad \Gamma \longrightarrow \Delta \\
\hline
\Gamma, a^n \longrightarrow \Delta, a^n
\end{array}
\quad
\frac{\Gamma \longrightarrow \Delta}{\Gamma \longrightarrow \Delta, a^n}
\quad
\frac{\Gamma \longrightarrow \Delta}{\Gamma, T^n \longrightarrow \Delta, T^n}
\quad
\frac{\Gamma \longrightarrow \Delta}{\Gamma, \hat{\alpha}^n \longrightarrow \Delta, \hat{\alpha}^n}
\quad
\frac{\Gamma \longrightarrow \Delta}{\Gamma \longrightarrow \Delta, \hat{\alpha}^n}$$

$$\frac{\Gamma[\hat{\alpha} := \tau] \longrightarrow \Delta[\hat{\alpha} := \tau]}{\Gamma, \hat{\alpha}^n \longrightarrow \Delta, \hat{\alpha}^n = \tau}
\quad
\frac{\Gamma[\hat{\alpha} := \tau] \longrightarrow \Delta[\hat{\alpha} := \tau'] \quad [\Delta]\tau = [\Delta]\tau'}{\Gamma, \hat{\alpha}^n = \tau \longrightarrow \Delta, \hat{\alpha}^n = \tau'}
\quad
\frac{\Gamma \longrightarrow \Delta[\hat{\alpha} := \tau]}{\Gamma \longrightarrow \Delta, \hat{\alpha}^n = \tau}$$

Fig. 10. Context extension

6.3 Soundness

We prove that the algorithm is sound and complete (§6.4) with respect to the declarative system. We start with definitions for reasoning about contexts.

Context definitions. Fig. 9 defines well-formedness of contexts. The judgment $\Gamma \vdash^n \Sigma$ states that the term context Σ is well-formed under the algorithmic context Γ at level n . The judgment ensures that all types in Σ are well-formed under Γ at level n .

The judgment $\Gamma \vdash^n \Delta$ states that Δ is well-formed under Γ at level n , ensuring that all variables in Δ have levels no greater than n . The only interesting case is the last rule, which checks that τ is well-formed at level m , with $m \leq n$. Additionally, the rule checks that $\hat{\alpha}$ is not free in $[\Delta]\tau$. Lastly, it also requires Δ to be well-formed. Intuitively, we need Γ as Δ may still refer to $\hat{\alpha}$.

We write $\Gamma \vdash^n \Gamma$, or often just Γ^n , to denote that a context Γ is well-formed under itself at level n . When the level does not matter, we also write Γ^∞ to mean that Γ is well-formed at some level.

Fig. 10 defines *context extension*, where the judgment $\Gamma \longrightarrow \Delta$ states that Γ is extended by Δ . Intuitively, context extension expresses a form of information increase, where Δ may contain more variables or solutions for existing variables. The last three rules substitute the solution for $\hat{\alpha}$ in the rest of the contexts, since the contexts may still refer to $\hat{\alpha}$.

Soundness. We now establish soundness, starting from soundness of promotion. Notably, since Ω is a complete context with all unification variables resolved, $[\Omega]\sigma$ produces a declarative type for any well-formed type σ .

Lemma 6.1 (Soundness of promotion). *If Γ^∞ and $\Gamma \vdash^n \sigma$ and $\Delta \longrightarrow \Omega$ and Ω^∞ , we have:*

- (1) if $\Gamma \vdash \sigma \rightsquigarrow_m^+ \tau \vdash \Delta$, then $\vdash^m [\Omega]\sigma <: [\Omega]\tau$.
- (2) if $\Gamma \vdash \sigma \rightsquigarrow_m^- \tau \vdash \Delta$, then $\vdash^m [\Omega]\tau <: [\Omega]\sigma$;

The lemma captures the essence of promotion: promoting a polymorphic type σ under positive polarity produces a supertype of σ , while promoting it under negative polarity produces a subtype. With that, we prove the soundness of subtyping:

Theorem 6.2 (Soundness of subtyping). *If Γ^∞ and Δ^∞ and $\Gamma \vdash^n \sigma_1$ and $\Gamma \vdash^n \sigma_2$ and $\Gamma \vdash^n \sigma_1 <: \sigma_2 \dashv \Delta$, where $\Delta \longrightarrow \Omega$ and Ω^∞ , then $\vdash^n [\Omega]\sigma_1 <: [\Omega]\sigma_2$.*

Lastly, we prove the soundness of typing, where we extend context application to term contexts, and thus $[\Omega]\Sigma$ produces a declarative context, and the notation $[\Omega]$ extracts all the type constructors from Ω and they are implicitly mapped to their levels in the declarative typing judgments.⁶

Theorem 6.3 (Soundness of typing). *Given $\Delta \longrightarrow \Omega$, where $\Gamma^\infty, \Delta^\infty, \Omega^\infty$, and $\text{fT}(\Gamma) \leq n$:*

(Inference) If $\Gamma \vdash^n \Sigma$ and $\Gamma \vdash \Sigma \vdash^n e \Rightarrow \sigma \dashv \Delta$ then $[\Omega]\Sigma, [\Omega]\vdash^n e \Rightarrow [\Omega]\sigma$.

(Checking) If $\Gamma \vdash^n \Sigma$ and $\Gamma \vdash \Sigma \vdash^n e \Leftarrow \sigma \dashv \Delta$ then $[\Omega]\Sigma, [\Omega]\vdash^n e \Leftarrow [\Omega]\sigma$.

6.4 Completeness

We now move to completeness. We start with completeness of promotion.

Lemma 6.4 (Completeness of promotion). *Given $\Gamma \longrightarrow \Omega$ and Γ^∞ and Ω^∞ and $\Gamma \vdash^m \tau$ and $\Gamma \vdash^n \sigma$:*

- *if $\vdash^n [\Omega]\tau <: [\Omega]\sigma$, then there exist Δ and Ω' such that $\Delta \longrightarrow \Omega'$ and $\Omega \longrightarrow \Omega'$ and $\Gamma \vdash \sigma \rightsquigarrow_m^- \tau' \dashv \Delta$ where $[\Omega']\tau' = [\Omega']\tau$.*
- *if $\vdash^n [\Omega]\sigma <: [\Omega]\tau$, then there exist Δ and Ω' such that $\Delta \longrightarrow \Omega'$ and $\Omega \longrightarrow \Omega'$ and $\Gamma \vdash \sigma \rightsquigarrow_m^+ \tau' \dashv \Delta$ where $[\Omega']\tau' = [\Omega']\tau$.*

Note that Ω and Δ may contain different but equivalent solutions for unification variables, such as $\Omega = (\hat{\alpha}^0 = \text{Int} \rightarrow \text{Int})$ and $\Delta = (\hat{\alpha}^0 = \hat{\beta} \rightarrow \hat{\beta}, \hat{\beta}^0 = \text{Int})$. Therefore, we show that there is a context Ω' that extends both Δ and Ω . The lemma states that subtyping between a polymorphic type and a monotone can be resolved by promoting the polymorphic type to τ' , with $[\Omega']\tau' = [\Omega']\tau$.

We proceed to completeness of subtyping and typing:

Theorem 6.5 (Completeness of subtyping). *Given $\Gamma \longrightarrow \Omega$, and Γ^∞ and Ω^∞ and $\Gamma \vdash^n \sigma_1$ and $\Gamma \vdash^n \sigma_2$, if $\vdash^n [\Omega]\sigma_1 <: [\Omega]\sigma_2$, then there exist Δ and Ω' such that $\Delta \longrightarrow \Omega'$ and $\Omega \longrightarrow \Omega'$ and $\Gamma \vdash^n [\Gamma]\sigma_1 <: [\Gamma]\sigma_2 \dashv \Delta$.*

Theorem 6.6 (Completeness of typing). *Given $\Gamma \longrightarrow \Omega$ and Γ^∞ and Ω^∞ and $\Gamma \vdash^n \Sigma$ and $\text{fT}(\Gamma) \leq n$:*

(Inference) If $[\Omega]\Sigma, [\Omega]\vdash^n e \Rightarrow \sigma$ and $\vdash^n [\Omega]\Sigma' <: [\Omega]\Sigma$ and $\Gamma \vdash^n \Sigma'$, then there exist Δ, Ω' , and σ' such that $\Delta \longrightarrow \Omega'$ and $\Omega \longrightarrow \Omega'$ and $\Gamma \vdash \Sigma' \vdash^n e \Rightarrow \sigma' \dashv \Delta$ and $\vdash^n [\Omega']\sigma' <: \sigma$.

(Checking) If $[\Omega]\Sigma, [\Omega]\vdash^n e \Leftarrow [\Omega]\sigma$ and $\Gamma \vdash^n \Sigma$, then there exist Δ and Ω' such that $\Delta \longrightarrow \Omega'$ and $\Omega \longrightarrow \Omega'$ and $\Gamma \vdash \Sigma \vdash^n e \Leftarrow [\Gamma]\sigma \dashv \Delta$.

Notably, completeness of inference allows algorithmic typing to produce a more general type than the declarative system, since the declarative system may not always generalize a let binding (§4.2). Thus, the theorem uses notion of *context subtyping*, denoted as $\vdash^n \Psi_1 <: \Psi_2$, where Ψ_1 assigns more general types to the same binding compared to Ψ_2 , and the inferred type is also more general.

7 Implementation

We have implemented level-based type inference for the Koka language [Leijen 2013], and included the modified Koka compiler in the artifact of this paper [Fan et al. 2025].

⁶We additionally require the type constructors appearing free in Γ to have levels no greater than the typing level.

Compiler implementation. Koka supports both let generalization and higher-rank polymorphism. Following the traditional approach, the existing implementation traverses the entire typing context to collect free type variables for generalization, and includes additional checks for skolem escape.

We implemented level-based type inference in a Koka compiler. Following the formalism, we associate each unification and skolem variable with a level, and keep track of the current level throughout type inference. The typing levels are incremented upon entering a new polymorphism scope and decremented when exiting it. This eliminates the need for context traversal during generalization. Similarly, skolemization happens at the incremented level when a lambda is checked against a propagated polymorphic type or when a type is checked to subsume another polymorphic type. The promotion process rejects the program if a skolem attempts to leak to a lower level.

We note that Koka has a polymorphic type-and-effect system with algebraic effect handlers [Plotkin and Power 2001; Plotkin and Pretnar 2009] and mutable reference cells. Our level-based generalization naturally supports effect polymorphism. In particular, generalization happens when typing named functions and top-level bindings that are total (akin to the *value restriction* [Wright 1995]).

Additionally, Koka supports *impredicativity* [Leijen 2008], where type variables can be instantiated with polymorphic types. As a result, the promotion implementation for Koka can produce a polymorphic type and does not require the polarity, as shown by the rule on the right. For example, promoting $\forall a. a \rightarrow a$ produces the type itself. In Koka, this is implemented by treating a as a bound variable without a level, rather than a skolem variable.

$$\frac{\Gamma, a^0 \vdash \sigma \rightsquigarrow_n \sigma'}{\Gamma \vdash \forall a. \sigma \rightsquigarrow_n \forall a. \sigma'}$$

Validation. To validate the implementation, we have run the modified compiler on the entire Koka test suite which includes 308 positive and negative tests. Our implementation produced results identical to the original compiler for 275 tests.

For the remaining tests, the modified compiler produced equivalent results after alpha-renaming of bound variables for 19 tests, where alpha-equivalence is needed because the constraint solver in the modified compiler generates different numbers of variable identifiers, which then appear in the generated core programs. Both compilers correctly rejected 7 negative tests (involving issues like skolem escapes). The modified implementation produced different error messages, as promotion detected skolem escapes earlier than the traditional context traversal approach. The remaining 7 tests involve an analysis to remove tail effect variables. The analysis is known to be fragile [Ikemori et al. 2022, §4.5], where the typability of a program is sensitive to small program transformations (specifically, lifting a term to a let binding influences typability). We leave developing a more robust analysis to future work.

Evaluation of generalization. It is clear that level-based generalization is computationally more efficient, as it does not involve traversing the entire typing context. Rémy [1992] introduced level-based generalization as “a simple and efficient presentation of ML type system”. Nevertheless, Rémy [1992], and subsequent work such as Kuan and MacQueen [2007], did not provide an evaluation.

We evaluated our level-based implementation in the Koka compiler against the original one, focusing on performance gains of level-based generalization in a relatively modern type-checker. To this end, we generated programs that stress the generalization process. Specifically, these generated programs have 200 simple functions nested within a top-level function with varying numbers of parameters. This structure models scenarios where a function relies on numerous local functions. Each nested function simply takes three parameters and returns one parameter from the top-level function. As a result, a program runs generalization 200 times, in a typing context whose size is proportional to the sum of the number of parameters in the top-level function and the number of functions already type-checked. The evaluation was performed on a MacBook Pro 2023 with 8-Core 64-bit Apple M3 CPU and 24 GB unified memory.

We present the evaluation results in Fig. 11, comparing Level Koka, the level-based implementation, with Koka the original compiler, where let generalization and skolem escape detection traverse the typing context for free type variables. We disabled tail effect removal for both compilers, ensuring identical typing results, so that the performance difference is due to the generalization strategies and the promotion overhead. We report the average type-checking time in milliseconds (ms) over 10 runs for each program. The results show that generalization in Level Koka is 2.9-3.7x faster than Koka on the programs. Moreover, Koka gets slower as the number of parameters in the top-level function grows, leading to a larger typing context.

It is important to note that these benchmark programs are specifically designed to stress the generalization process, and the observed performance is specific to the data structures used within the Koka type checker. In practice, the typing context may not reach the size of 2000, and the overall running time of a compiler is impacted by various other phases beyond type checking. We interpret the evaluation results as preliminary empirical evidence supporting the folklore that level-based generalization is more efficient. In the future we are interested in studying the performance impact on larger-scale Koka applications.

8 Language Extensions

We explore related language extensions and discuss how modern type checkers, specifically the Glasgow Haskell Compiler (GHC) and the OCaml type checker, use levels in their implementations.

Kind polymorphism. While type variables in this paper all have the same kind (i.e. the kind $\star$), modern type checkers often employ higher kinds or *kind polymorphism* [Yorgey et al. 2012]. With kind polymorphism, the kind of a type variable can include a kind variable. Extending levels to support kind variables is relatively straightforward: each kind variable is associated with a level, and promoting a type variable also promotes its kind. Xie et al. [2019] provide a detailed formalism of kind inference in the setting of ordered contexts [Dunfield and Krishnaswami 2013].

GADTs. Similar to the formalism presented in this paper, GHC associates each type variable with a level and uses levels for generalization and skolem escape check. Additionally, GHC uses levels when type-checking programs with *generalized algebraic datatypes* (GADTs).

Specifically, consider the example on the right taken from Vytiniotis et al. [2011]. We can type *test* with either of the following two types that are not a subtype of each other: (1) $\forall a. T\ a \rightarrow \text{Bool} \rightarrow \text{Bool}$; (2) $\forall a. T\ a \rightarrow a \rightarrow a$. This example demonstrates the known issue that type inference for GADTs does not always have principal types [Cheney and Hinze 2003; Vytiniotis et al. 2011]. GHC rejects *test* using the concept of *untouchability*. Specifically, if the return type of *test* is a unification variable $\hat{a}$, this variable is considered untouchable (i.e. cannot be solved) under the local assumption $a \sim \text{Bool}$ where $T1\ n$ has type $T\ a$ in the first case, as $\hat{a}$ could have two incompatible solutions: $\hat{a} = a$, or $\hat{a} = \text{Bool}$.

GHC implements untouchability using levels. Specifically, GHC's type inference is based on constraint generation and solving. In the above example, the GADT match introduces an *implication constraint* $a \sim \text{Bool} \Rightarrow \hat{a} \sim \text{Bool}$. Importantly, GHC increments the level when checking a GADT match, and associates the implication constraint with the incremented level (say 2). Since $\hat{a}$ has a

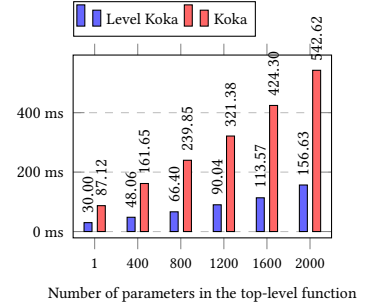


Fig. 11. Evaluation

lower level (say 1) and is under a local assumption, it is considered untouchable, as unifying $\hat{\alpha}$ may not produce principal types. This mechanism prevents solving $\hat{\alpha}$ with *Bool*.

We remark that using levels for untouchable variables shares similarities with skolem escape checks. In both cases, levels indicate the valid scope of a unification variable: it prevents unification with a skolem variable of a higher level, or within an implication constraint that has a higher level.

Type families. GHC also supports *type families* [Eisenberg et al. 2014; Stolarek et al. 2015]. Recall that when unifying $\hat{\alpha}^1$ with *Maybe* $\hat{\beta}^2$, we promote $\hat{\beta}^2$ to level 1. Interestingly, given a type family F , unifying $\hat{\alpha}^1$ with $F \hat{\beta}^2$ should not promote $\hat{\beta}^2$, as $F \hat{\beta}^2$ can potentially reduce to, say, *Int*, which is well-formed at level 1. As a result, GHC does not promote variables under a type-family application.

The OCaml type checker. Levels are also used in the OCaml type checker, as detailed in a blog post by Kiselyov [2022]. While promotion in our system and GHC involves creating new unification variables, levels in OCaml use mutable references and can thus be updated in-place.

OCaml prevents local definitions from leaking. For example, the program with local modules on the right does not type-check. OCaml achieves that by incrementing the typing level when type-checking a local module, and later checking that the level of the result type has the original level.

Interestingly, in OCaml, every type is associated with a level, maintained during unification. The design enables efficient level access through a constant-time lookup. OCaml thus employs several techniques to optimizing level-related operations. For example, during generalizing, if a type's level is not greater than the current typing level, the type checker doesn't need to traverse that type's structure. OCaml also adjusts levels to relax the value restriction [Garrigue 2004], by lowering the level of type variables appearing in contravariant positions, preventing their generalization.

OCaml also supports GADTs using the concept of *ambivalent types* [Garrigue and Rémy 2013]. More concretely, types in OCaml carry an additional *scope*. Ensuring that an ambivalent type does not escape its scope is equivalent to checking if its scope is no greater than its level.

Another interesting use of levels is that OCaml associates bound variables with a very large level (10^8). Thus, instantiating can skip types without such a level as they have no bound variables.

```
let y =
  let module M =
    struct
      type t = Foo
      let x = Foo
    end
  in M.x
```

9 Related Work and Conclusion

We have discussed most related work on levels throughout the paper [Kiselyov 2022; Kuan and MacQueen 2007; Rémy 1992]. *Ordered contexts* [Dunfield and Krishnaswami 2013; Gundry et al. 2010] is an approach adopted in several subsequent works [Dunfield and Krishnaswami 2019; Xie et al. 2019; Zhao et al. 2019]. While ordered contexts offer an elegant framework for reasoning about type inference, they focus more on theoretical foundations than practical implementations. Xie [2021, §6.5] informally compared order contexts with levels for higher-rank polymorphism.

This work seems to be the first comprehensive formalism of level-based type inference beyond let generalization. While we explored a range of language features implemented using levels, our investigation is not exhaustive. We are interested in extending the formalism with more features, such as GADTs. Furthermore, following Garrigue [2015]; Zhao et al. [2018, 2019], we would like to mechanize the proofs for our algorithmic system in the future.

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Data Availability Statement

The Coq mechanization of the declarative systems and our implementation of the level-based type inference for the Koka compiler are included in the artifact of this paper [Fan et al. 2025].

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